

CLOUD COMPUTING

SAAS DEVELOPMENT ASSESSMENT

Assignment Title

SaaS Web-Based Point-of-Sale (PoS) Application Development and Demonstration

1. Overview

In this part of the assessment, students are required to design and develop a **web-based Point-of-Sale (PoS) application** that operates as a **Software-as-a-Service (SaaS)** solution.

The system must support **multi-tenant usage**, allowing multiple independent tenants (e.g., shops or companies) to use the same application while keeping their data isolated.

Note:

A *tenant* refers to a **shop or a company** that uses the system. Each tenant may have **multiple users (e.g., staff members)** who access the application under the same tenant account.

2. Objective

- Develop a functional web-based PoS system
- Implement core business logic for sales transactions
- Support multi-tenant data handling

- Integrate email functionality for receipt delivery
 - Demonstrate a working SaaS application
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3. Functional Requirements

3.1 Multi-Tenant Support (Mandatory)

The application must support **multi-tenancy**, meaning:

- Each tenant represents a separate shop or company
- Each tenant can have multiple users
- Data must be logically isolated between tenants
- Users must only access data belonging to their tenant

Minimum implementation:

- Include a **Tenant ID** in all relevant data (e.g., products, transactions)
 - Ensure all queries are filtered by Tenant ID
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3.2 Product Management (Simplified)

Create and manage products per tenant

Each product must include:

- Product name
- Price
- (Optional) Category or description

3.3 Sales Transaction Processing

Create a new transaction

Add multiple products to a transaction

Specify quantity for each product

Automatically calculate:

- Subtotal
- Total amount

3.4 Receipt Generation

Generate a digital receipt for each transaction

Receipt must include:

- Transaction ID
 - Date and time
 - List of purchased items
 - Quantity and price
 - Total amount
 - Tenant/store identification
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3.5 Email Receipt Sending

- Allow user to enter a customer email address
 - Send receipt via email after transaction completion
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4. Technical Requirements

- Web-based application
 - Programming language/framework: Open (lecturer approval required)
 - Must be accessible via a public URL
 - Must support **multi-tenant data handling**
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5. Demonstration Requirements

Students must demonstrate:

- Functional PoS application
 - Multi-tenant behavior (e.g., different tenants see different data)
 - Product and transaction management per tenant
 - Email receipt sending
 - Application running from a cloud-hosted environment
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6. Assessment Criteria (100 points)

Criteria	Points
Application Functionality	30

Multi-Tenant Implementation	20
UI/UX and Usability	15
Code Quality & Structure	10
Email Integration (Working)	10
Live Demonstration	15
Total	100

7. Submission Requirements

- Source code (GitHub or equivalent)
 - Working application deployed online
 - Demonstration during scheduled session
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